The Projected Predictive Route: Theory and Computational Path

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Abstract

This note describes, from encoding to loss, the computation executed by the `vfe3` language model under `attention_mode="projected_predictive"`, the route behind the current best results. Each token is an agent with a Gaussian state and an invertible frame. A layer measures query and key laws in a comparison space, routes by a Gibbs distribution over their divergences, and transmits content through independent sending and receiving maps whose composition is the matrix link $L_{ij}^{(h)} = U_i O_h T_h U_j^{-1}$. We state which parts are exact consequences of a declared model and which are architectural choices. The note documents a new option, `predictive_covariance="moment_matched"`, which replaces the covariance passthrough of the evaluated route by the exact moment-matched covariance of the same predictive kernel whose mean the layer already computes. With it, the belief covariance becomes context dependent. We give its derivation, its reductions, its covariance under frame changes, its projection onto diagonal families, and its implementation and tests, and we list what remains untested.

1 Scope

The GL(K) attention manuscript develops the construction in full generality (Dennis, 2026a). This note follows one route through it: the configuration family selected by `attention_mode="projected_predictive"`, in the order in which the code executes it. Section 2 fixes the states and frames. Sections 3–5 describe one communication layer: comparison and routing, content, and the covariance. Section 6 covers the seams between layers, Section 7 the decoder and objective, and Section 8 maps every configuration toggle to the equation it controls. Section 9 separates exact statements from declared choices and records the evidence.

2 States, frames, and encoding

Token position i carries a Gaussian state in local coordinates on a fiber $E_i \cong \mathbb{R}^K$ and an invertible frame U_i identifying the reference space $\mathsf{V} = \mathbb{R}^K$ with E_i:

$$q_i = \mathcal{N}(\mu_i, \Sigma_i), \qquad \xi_i = U_i^{-1}\mu_i, \qquad C_i = U_i^{-1}\Sigma_i U_i^{-\top}. \tag{1}$$

The pair (ξ_i, C_i) is the state in reference coordinates. A passive rechoice of agent i's frame acts by

$$(\mu_i, \Sigma_i, U_i) \mapsto (g_i\mu_i, g_i\Sigma_i g_i^{\top}, g_i U_i), \qquad g_i \in \mathrm{GL}(K), \tag{2}$$

and leaves (ξ_i, C_i) unchanged. Every routing cost and message below is computed from reference moments and is therefore invariant, or covariant, under Eq. (2).

Encoding. Each token type c has a prior bank entry $(\mu_p^c, \Sigma_p^c, \phi_c)$: a Gaussian embedding in the sense of Vilnis and McCallum (2015) with a Lie-algebra frame coefficient (Dennis, 2026a). Under `encode_mode="per_token"` the input state is the prior of the token, $q_i^{(0)} = p_{x_i}$. The frame is a product of token and positional factors, $U_i = \exp(X_{x_i}) \exp(Y_i)$, with Y_i learned when `pos_phi="learned"`. Under `gauge_group="block_glk"` both factors are block diagonal with one $d \times d$ block per head, $K = Hd$. The projected route requires a flat frame that does not change within a forward pass (`e_phi_lr=0`, `n_e_steps=1`), so the frame is built once per forward pass and shared by all layers.

3 One layer: comparison and routing

A layer has H heads. Head h owns comparison maps $S_{Q,h}, S_{K,h} : \mathsf{V} \to \mathbb{R}^r$ of full row rank and content maps $T_h : \mathsf{V} \to \mathbb{R}^s$ and $O_h : \mathbb{R}^s \to \mathsf{V}$. In the code $r = s = d$. Under `relational_input_scope="head"` the maps read only head block h of V, and under `"full"` they read all K coordinates, as learned projections do in conventional attention (Vaswani et al., 2017). Under `relational_output_scope="head"` O_h writes only block h, and under `"full"` it writes all of V.

Comparison laws. The query and key laws are Gaussian pushforwards of the reference states into the comparison space,

$$\begin{aligned} q_{Q,i}^h &= \mathcal{N}\big(a_i, V_{Q,i}\big), \quad a_i = S_{Q,h}\xi_i, \quad V_{Q,i} = S_{Q,h} C_i S_{Q,h}^\top, \\ p_{K,j}^h &= \mathcal{N}\big(b_j, V_{K,j}\big), \quad b_j = S_{K,h}\xi_j, \quad V_{K,j} = S_{K,h} C_j S_{K,h}^\top. \end{aligned} \tag{3}$$

Both reduced covariances are positive definite because $C_i \succ 0$ and the maps have full row rank. The routing cost under `relational_routing="gaussian_kl"` is the exact Gaussian divergence

$$\begin{aligned} D_{ij}^h = D_{\mathrm{KL}}\big(q_{Q,i}^h \| p_{K,j}^h\big) = \tfrac{1}{2}\big[\, &\mathrm{Tr}(P_j V_{Q,i}) + (a_i - b_j)^\top P_j (a_i - b_j) - r \\ &+ \log\det V_{K,j} - \log\det V_{Q,i}\big], \qquad P_j = V_{K,j}^{-1}. \end{aligned} \tag{4}$$

Its mean compatibility contains the bilinear term $a_i^\top P_j b_j$, the covariance-weighted analogue of the dot-product score. The terms in $\mathrm{Tr}(P_j V_{Q,i})$ and $\log\det V_{K,j}$ carry covariance information and are pair dependent whenever the key covariances differ across sources. Under `relational_routing="noisy_query"` the cost is instead $-\log \mathcal{N}(a_i; b_j, V_{Q,i} + V_{K,j})$, the expected-likelihood cost under which query uncertainty tempers the row (Dennis, 2026a).

Gibbs routing. With a source prior π_{ij}^h and a temperature $\tau_h > 0$, the routing weights minimize $\sum_j \beta_j D_{ij}^h + \tau_h \sum_j \beta_j \log(\beta_j / \pi_{ij}^h)$ on the simplex, with unique solution

$$\beta_{ij}^h = \mathrm{softmax}_j\big(-D_{ij}^h/\tau_h + \log \pi_{ij}^h\big). \tag{5}$$

The causal support sets $\pi_{ij}^h = 0$ for $j > i$. Under `beta_attention_prior="causal_alibi"` the prior adds a head-dependent linear distance penalty. The temperature is $\tau_h = \kappa\sqrt{d}$ with $\kappa =$ `kappa_beta`. Equation (5) is an exact posterior at every temperature: attaching to each receiver and head a binary compatibility record with likelihood $\exp(-D_{ij}^h/\tau_h)$ makes β_{ij}^h the posterior of the source label given the record (Dennis, 2026a). This establishes consistency with exact inference; it does not show that attention must be inference.

4 One layer: content and the predictive mean

The content maps compose with the frames into the matrix link

$$L_{ij}^h = U_i M_h U_j^{-1}, \qquad M_h = O_h T_h, \tag{6}$$

which obeys the endpoint law $L_{ij}^h \mapsto g_i L_{ij}^h g_j^{-1}$ under Eq. (2). The predictive message and the layer's mean update are

$$m_i = \sum_h \sum_j \beta_{ij}^h L_{ij}^h \mu_j = U_i \sum_h O_h \sum_j \beta_{ij}^h T_h \xi_j, \qquad \mu_i^+ = r\mu_i + m_i, \tag{7}$$

with $r = 1$ under `attention_residual=True` and $r = 0$ otherwise. The message is linear in the source means; the only state nonlinearity in the layer is the dependence of β on the states through Eq. (4). At initialization T_h and O_h are identity blocks placed on head h, scaled by $1/\sqrt{2L}$ under `relational_output_init="depth_scaled"`.

The predictive kernel. Equation (7) is the mean of a declared kernel. Give each receiver and head an independent source label $J^h \sim \beta_i^h$ and take

$$k_i' \mid k, J \sim \mathcal{N}\Big(r k_i + \sum_h L_{iJ^h}^h k_{J^h},\ Q_i\Big), \tag{8}$$

with source states $k_j \sim \mathcal{N}(\mu_j, \Sigma_j)$ independent across j and the self edge included. At $k_j = \mu_j$ its mean is μ_i^+. The kernel leaves T_h, O_h, and Q_i free and declares the unit persistence coefficient, additivity across heads, and one label per head for both routing and content. These are modeling choices recorded as such.

5 One layer: the covariance

5.1 Passthrough

Under `predictive_covariance="passthrough"`, the default and the setting of all evaluated runs, the layer returns $\Sigma_i^+ = \Sigma_i$. The covariance is then a routing state: it enters Eq. (4) and the decoder, but it is never updated after encoding. Because the encoded covariance is the lookup $\Sigma_p^{x_i}$, every downstream covariance is a function of the token identity alone. The calibration probe saved by the runs records this: its covariance source is labeled input-token variance, and in two recent $K = 64$ runs the across-token coefficient of variation of $\operatorname{Tr}\Sigma$ was 0.02 to 0.03.

5.2 Moment matching

Proposition 1 (Moment-matched predictive covariance)**.** *Under the kernel of Eq. (8), the covariance of k_i' in reference coordinates is*

$$\begin{aligned} C_i^{\mathrm{att}} = \widetilde{Q}_i + r\Big[C_i + \sum_h \beta_{ii}^h \big(M_h C_i + C_i M_h^\top\big)\Big] + \sum_j \sum_{h,h'} c_{ij}^{hh'} M_h C_j M_{h'}^\top \\ + \sum_h \Big[\sum_j \beta_{ij}^h n_{ij}^h n_{ij}^{h\top} - \bar{n}_i^h \bar{n}_i^{h\top}\Big], \end{aligned} \tag{9}$$

where $\widetilde{Q}_i = U_i^{-1} Q_i U_i^{-\top}$, $c_{ij}^{hh} = \beta_{ij}^h$, $c_{ij}^{hh'} = \beta_{ij}^h \beta_{ij}^{h'}$ for $h \neq h'$, $n_{ij}^h = M_h \xi_j$, and $\bar{n}_i^h = \sum_j \beta_{ij}^h n_{ij}^h$. The local covariance is $\Sigma_i^+ = U_i C_i^{\mathrm{att}} U_i^\top$.

Proof. Write $X = rk_i + \sum_h L^h_{iJ^h} k_{J^h} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, Q_i)$. By the law of total covariance, $\operatorname{Cov} X = \mathbb{E} \operatorname{Cov}(X \mid J) + \operatorname{Cov} \mathbb{E}(X \mid J) + Q_i$. Given the labels, $X - \varepsilon = \sum_s A_s k_s$ with $A_i = rI + \sum_h \mathbf{1}[J^h = i] L^h_{ii}$ and $A_s = \sum_h \mathbf{1}[J^h = s] L^h_{is}$ for $s \neq i$, so $\operatorname{Cov}(X \mid J) = \sum_s A_s \Sigma_s A_s^\top + Q_i$ by independence of the sources. Averaging over the independent labels uses $\mathbb{E}(\mathbf{1}[J^h = s]\mathbf{1}[J^{h'} = s]) = c^{hh'}_{is}$ and gives the terms in C_i, β^h_{ii}, and $c^{hh'}_{ij}$. For the second term, $\mathbb{E}(X \mid J) = r\mu_i + \sum_h L^h_{iJ^h} \mu_{J^h}$ is a sum of independent head contributions, so its covariance is the sum of the per-head spreads. Conjugating by U_i^{-1} removes every frame, since $L^h_{ij} \Sigma_j L^{h'\top}_{ij} = U_i M_h C_j M_{h'}^\top U_i^\top$. □

The proposition is item (2) of the exact-status theorem of the GL(K) manuscript (Dennis, 2026a), which already observes that passthrough retains only the term C_i. Under `predictive_covariance="moment_matched"` the layer returns Σ_i^+ with the reference floor

$$\widetilde{Q}_i = \epsilon \frac{\operatorname{Tr} C_i}{K} I, \qquad Q_i = \epsilon \frac{\operatorname{Tr} C_i}{K} U_i U_i^\top, \qquad \epsilon = \texttt{predictive_covariance_floor}. \tag{10}$$

The trace $\operatorname{Tr} C_i$ is invariant under Eq. (2), so Q_i transforms as a covariance. The kernel leaves $\widetilde{Q}_i$ free; this choice is made for conditioning. A relative floor ϵC_i would compound along a residual-free stack: without the persistence term, the only lower bound it supplies after ℓ layers is ϵ^ℓ times the encoded covariance, and a configuration sweep with rank-one content maps drove the condition number past 10^{10} and lost float32 definiteness at the second layer. The trace-scaled floor bounds the reference condition number by the ratio of the largest content variance to $\epsilon \operatorname{Tr} C_i / K$ at every depth.

Properties. Four consequences follow directly. First, $C_i^{\text{att}} - \widetilde{Q}_i$ is a covariance and hence positive semidefinite, so $\Sigma_i^+ \succeq Q_i \succ 0$: the output is positive definite whatever the routing and whatever the rank of M_h. Second, Σ_i^+ obeys the passive law $\Sigma_i^+ \mapsto g_i \Sigma_i^+ g_i^\top$, because C_i^{att} depends only on reference moments and the invariant weights, and the floor of Eq. (10) transforms in the same way. Third, the between-source spread depends on the source means through β and n^h_{ij}, so the covariance becomes context dependent: it grows when a receiver's attended sources carry disagreeing content. Fourth, the cross-head terms $h \neq h'$ are nonzero whenever two heads route to a common source and $T_h C_j T_{h'}^\top \neq 0$, which occurs for a full covariance or under full input scope. An earlier plan that dropped these terms as an independent-heads approximation is superseded by Eq. (9).

Reductions. With the residual off, $C_i^{\text{att}} = \operatorname{Cov}(m_i) + \widetilde{Q}_i$ in reference coordinates, the predictive reading in which the message replaces the state. With the residual on, the persistence term C_i and the self-edge cross terms appear. There is no monotonicity in depth in general: the cross terms $\beta^h_{ii}(M_h C_i + C_i M_h^\top)$ can be negative. If every $M_h = 0$, the output is $r\Sigma_i + Q_i$. If each head routes all weight to one source, the spread term vanishes and the cross-head weights reduce to indicators of a shared source.

Growth with depth under the residual. With the residual on and the identity initialization $M_h \approx$ head projectors summing to I, the first token routes only to itself, its kernel is $k_i' \approx 2k_i$, and its covariance quadruples at each layer, together with a doubling of its mean. A configuration sweep of tiny models observed maximal eigenvalues 4, 16, 64 over three layers, and extrapolation gives 256 at the launcher's four layers. This is the exact covariance of the declared kernel, not a numerical defect, and no ceiling is applied, since `sigma_max` bounds only the transported E-step and clamping would break the moment match. With the residual off, the observed trace stayed within 4 to 4.7 over three layers; `relational_output_init="depth_scaled"` halves the per-layer factor. Because the prior-bank decoder reads the final covariance, the residual setting now affects the decoder through the covariance scale as well as through the mean.

Diagonal families. For `gaussian_diagonal` and `gaussian_diagonal_exact`, the stored covariance is the diagonal of Σ_i^+. This is the $D_{\mathrm{KL}}(p\|q)$ moment projection onto diagonal Gaussians, which matches the marginal variances. For `gaussian_frame_diagonal`, the projected route supplies an intrinsic value frame (the identity, or a rotation under rotary position encoding), and the stored variances are the diagonal in that frame, the projection onto the family of covariances diagonal in the frame. For `gaussian_full`, Σ_i^+ is stored in full. Unlike passthrough, a full family then carries genuinely off-diagonal covariance even from a diagonal encoded table.

Value frame. Content and covariance use the same frame, the value frame, which coincides with the routing frame except under rotary position encoding with `rope_on_value=False` and under the frame-diagonal family.

Computation. The implementation (`RelationalOperators.predictive_covariance_update`) forms the reference covariances C_j, pushes them into the value space per head pair, $Y_j^{hh'} = \hat{T}_h C_j \hat{T}_{h'}^\top$, pools them with the weights $c_{ij}^{hh'}$, adds the pooled second moment of $v_j^h = \hat{T}_h \xi_j$ minus the outer product of its mean on the head diagonal, lifts by $\hat{O}_h$, adds the residual terms, and conjugates by the value frame. Here $\hat{T}_h \in \mathbb{R}^{d\times K}$ and $\hat{O}_h \in \mathbb{R}^{K\times d}$ embed the head-scope maps in the full state space, so every scope uses one code path. The pooling is an attention with $d \times d$ values per head pair, so the dominant costs are $O(HdK^2 + H^2d^2K)$ per token for the pushforward, $O(N^2H^2d^2)$ per sequence for the pooling, and $O(K^3)$ per token for the frame conjugations. All operations are differentiable, so gradients flow to the source means and covariances, the routing, the content maps, and the frame.

6 Between layers

After the attention step, a block applies optional seams in a fixed order: head mixer, cross-type coupling, block normalization, block MLP. The evaluated launcher uses none of the first three. The normalizations act on the mean only. The block MLP (`block_mlp_mode`) is a coordinate GELU MLP by default (Hendrycks and Gimpel, 2016). It passes the covariance through unless `block_mlp_covariance="delta_full"` applies the delta-method pushforward $J\Sigma J^\top$ with a covariant floor. Under `"gaussian_template"` it is replaced by template routing, a derived Gibbs readout of learned Gaussian templates that reads the diagonal of the covariance (Dennis, 2026b). With a moment-matched attention covariance, the template step therefore sees context-dependent uncertainty for the first time. The stack passes each block's output state to the next block and blends the state into the next prior by `prior_handoff_rho`. The predictive readout does not read the prior.

7 Decoder and objective

Under `decode_mode="linear"`, the next-token law is $\mathrm{softmax}_v(W_{\mathrm{out}}\mu_i^{(L)} + b + \kappa_{\mathrm{uni}} \log p_{\mathrm{uni}})$, a Gibbs readout of shared-covariance templates that reads the mean only. Under the prior-bank decoders (`diagonal_chunked`, `full_chunked`, `family_chunked`), the readout is

$$\hat{p}_i(v) \propto \omega_v \exp\big[-D_{\mathrm{KL}}(q_i^{(L)}\|p_v)/\tau_{\mathrm{dec}}\big], \tag{11}$$

whose exponent contains $\mathrm{Tr}(\Sigma_{p,v}^{-1}\Sigma_i^{(L)})$. The decoder therefore reads the final covariance. Under passthrough this is the input-token covariance; under moment matching it is the context-dependent

covariance of Proposition 1, so an uncertain final belief is scored against each class template with its full spread. Training minimizes next-token cross-entropy through every differentiable component. An optional M-step self-coupling term $\sum_i \alpha_i D_{\mathrm{KL}}(q_i^* \| p_i)$ is weighted by `mstep_self_coupling_weight`, which is zero in the evaluated launcher. No E-step gradient descent runs on this route: the layer update is the readout of Eqs. (7) and, optionally, (9).

8 Configuration map

Table 1 lists the toggles that shape this route, the equation each controls, and the value in the current launcher `train_vfe3.py`.

Toggle	Controls	Launcher
`attention_mode`	selects this route	`projected_predictive`
`family`	stored covariance layout and its projection	`gaussian_full`
`gauge_group`, `gauge_transport`	frame blocks, frame learned or ablated	`block_glk`, `on`
`pos_phi`, `pos_rotation`	positional frame factor, rotary encoding	`learned`, `none`
`relational_input_scope`	S and T read head block or all of V	`full`
`relational_output_scope`	O writes head block or all of V	`full`
`relational_routing`	cost of Eq. (4) or noisy query	`gaussian_kl`
`beta_attention_prior`, `kappa_beta`	π_{ij}^h and $\tau_h = \kappa\sqrt{d}$ in Eq. (5)	`causal_alibi`, 1
`attention_residual`	r in Eqs. (7) and (9)	`True`
`relational_output_init`	initial scale of O_h	`identity`
`predictive_covariance`	passthrough or Eq. (9)	`passthrough`
`predictive_covariance_floor`	ϵ in Eq. (10)	10^{-4}
`norm_type_block`, `norm_type_final`	mean normalization seams	`none`, `none`
`block_mlp_mode`, `block_mlp_covariance`	block MLP and its covariance contract	`coordinate`, `passthrough`
`decode_mode`, `decode_tau`	decoder of Eq. (11)	`full_chunked`, 0.1
`mstep_self_coupling_weight`	self-coupling term	0

Table 1: Configuration toggles of the projected predictive route. The `predictive_covariance` pair and `block_mlp_mode` are now written explicitly in the launcher at their defaults.

The computational path is: prior-bank encoding (`PriorBank.encode`); frame construction once per forward pass (`VFEModel._canonical_frame_context`); for each layer, `vfe_block` calls `e_step_iteration`, whose predictive branch calls `relational_predictive_readout`, which computes the costs (`relational_score`), the weights (`attention_weights`), the mean (`predictive_content`), and, under moment matching, the covariance (`predictive_covariance_update`); then the block seams; then the decoder and loss.

9 Status and evidence

Exact. The Gibbs weights are the unique minimizer of the routing objective and an exact posterior under the compatibility record. The predictive mean is the exact mean of the kernel of Eq. (8), and Proposition 1 gives its exact covariance. Every routing cost, message, and moment-matched

covariance transforms correctly under Eq. (2). The diagonal-family outputs are exact $D_{\mathrm{KL}}(p\|q)$ moment projections.

Declared. The kernel itself is a modeling choice: unit persistence, additive heads, independent head labels, independent sources, a single Gaussian collapse between layers (the true layer output is a mixture), and the floor of Eq. (10). The block MLP, normalization, and decoder choices are architectural. Moment matching makes the covariance consistent with the declared kernel; it does not make the stack as a whole the solution of one free-energy minimization.

Verified. The implementation agrees with an independent brute-force enumeration of every head-label tuple, computed in local coordinates from Eq. (8) rather than from Eq. (9), to 10^{-10} in float64. The check covers all combinations of input and output scope, residual, and compact or dense frames, and both diagonal and full covariances. Further tests cover the passive law, positive definiteness under sharp routing with zero content maps, finite-difference gradients with respect to the means, covariances, routing weights, and content maps, configuration validation and serialization, and forward and backward passes of two-layer models for every admissible family, with linear and prior-bank decoders. A float32 depth recursion with rank-one content maps and no residual stays positive definite, with a bounded condition number, over twelve layers; the same recursion with a relative floor fails at the second layer. The existing test files of the projected route pass unchanged, and the default path is untouched. A configuration sweep of roughly six thousand tiny-model combinations across routing costs, priors, temperatures, scopes, floors, head counts and depths, all admissible families, gauge and positional settings, the float64 frame path, block MLP modes, head mixer, normalizations, prior handoff, self-coupling, every decoder, the training loop, checkpoints, diagnostics, reporting, sampling, and CPU mixed precision found no configuration that runs under passthrough and fails under moment matching once the floor was changed. The overhead of the update on a CPU forward and backward pass at $K = 64$, $H = 2$, $N = 32$, $L = 2$ was about 1.5 times.

Untested. No training run uses the moment-matched covariance. Whether it improves predictive loss, whether the evolved covariance passes the calibration gate (across-token variation above 0.10 with a positive rank correlation to cross-entropy), and whether it helps the prior-bank decoder or template routing are open. The matched comparison against passthrough, with several seeds, is the experiment that decides.

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